

Develop linear regression models using the least square method to predict a patients maximum heart rate (thalach) from selected features. analyse prediction accuracy using MSE, and compare the models to determine which is more suitable which is more suitable for heart health risk estimation.

1. load the dataset
2. use age as independent variable and thalach as dependent variable
3. train model using linear regression
4. plot regression line and predict the points using matplotlib
5. compute and display slope and interscept,MSE

```
In [1]: import pandas as pd
```

```
In [3]: df=pd.read_csv(r"C:\Users\R3ai\Downloads\heart (1).csv")
```

```
In [4]: df
```

```
Out[4]:
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	c
0	63	1	3	145	233	1	0	150	0	2.3	0	
1	37	1	2	130	250	0	1	187	0	3.5	0	
2	41	0	1	130	204	0	0	172	0	1.4	2	
3	56	1	1	120	236	0	1	178	0	0.8	2	
4	57	0	0	120	354	0	1	163	1	0.6	2	
...	...	...	...	...	...	...	...	...	...	...	...	...
298	57	0	0	140	241	0	1	123	1	0.2	1	
299	45	1	3	110	264	0	1	132	0	1.2	1	
300	68	1	0	144	193	1	1	141	0	3.4	1	
301	57	1	0	130	131	0	1	115	1	1.2	1	
302	57	0	1	130	236	0	0	174	0	0.0	1	

303 rows × 14 columns

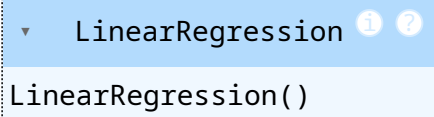
```
In [6]: from sklearn.linear_model import LinearRegression
import numpy as np
```

```
In [7]: x=df["age"].values.reshape(-1,1)
```

```
In [8]: y=df["thalach"]
```

```
In [9]: model=LinearRegression()
```

```
In [10]: model.fit(x,y)
```

Out[10]:  LinearRegression()
The image shows a dropdown menu for the 'LinearRegression' class. The menu is open, showing the class name 'LinearRegression' and its signature 'LinearRegression()'.

```
In [11]: a1=model.coef_[0]
```

```
In [12]: a0=model.intercept_
```

```
In [13]: print("a1(Slope):",round(a1,2))
```

a1(Slope): -1.01

```
In [14]: print("a0 (Intersept)",round(a0,2))
```

a0 (Intersept) 204.29

```
In [15]: from sklearn.metrics import mean_squared_error
```

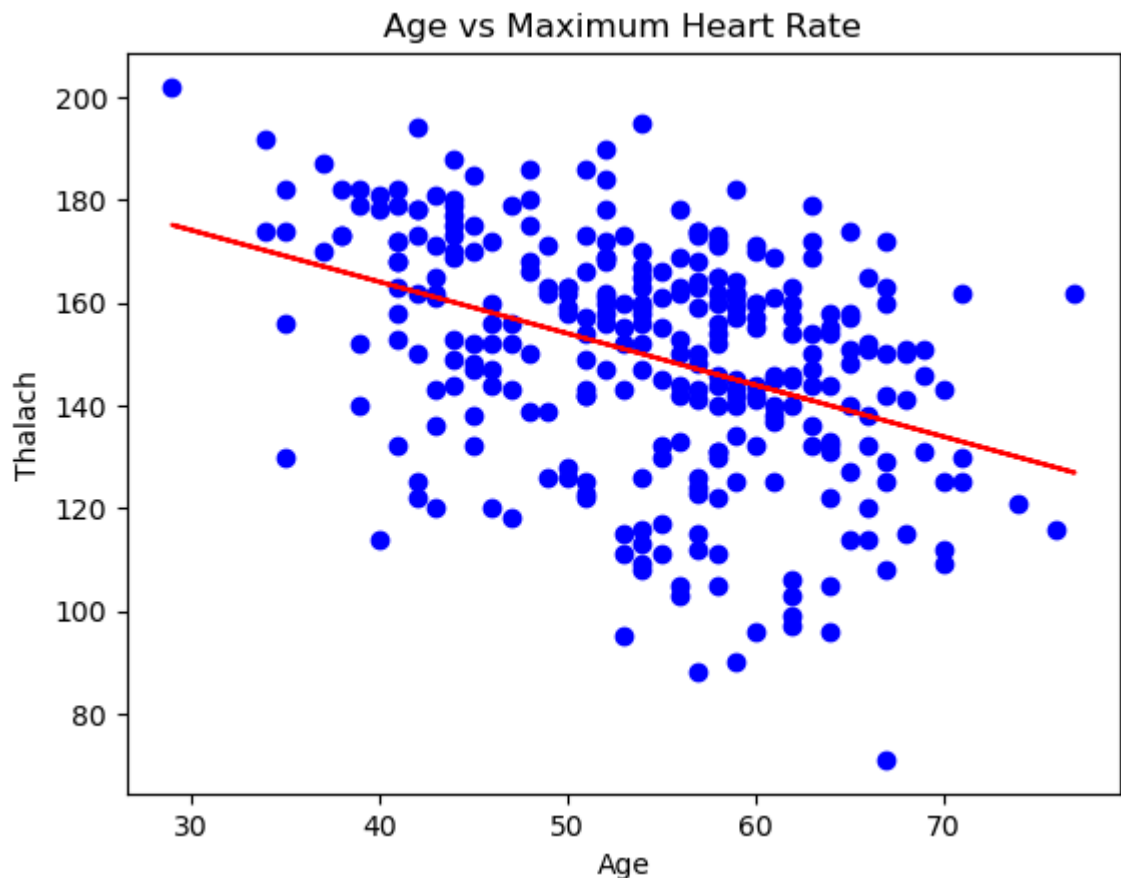
```
In [16]: y_pred_simple=model.predict(x)
```

```
In [17]: mse_simple=mean_squared_error(y,y_pred_simple)
```

```
In [18]: print("Mean Squared Error (MSE)= ",round(mse_simple,2))
```

Mean Squared Error (MSE)= 439.87

```
In [23]: import matplotlib.pyplot as plt
plt.scatter(x, y, color="blue")
plt.plot(x, y_pred_simple, color="red")
plt.xlabel("Age")
plt.ylabel("Thalach")
plt.title("Age vs Maximum Heart Rate")
plt.show()
```



Predict thalach using age, trestbps, chol as the independent variables and thalach as the dependent variable

1. train the multilinear regression model
2. find and display coefficients and interscepts, MSE
3. Do Preprocessing Also.

```
In [28]: import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

df = df.drop_duplicates()
df = df.dropna()

X = df[["age", "trestbps", "chol"]]
y = df["thalach"]

model = LinearRegression()
model.fit(X, y)

y_pred = model.predict(X)

print("Coefficients:", model.coef_)
print("Intercept:", model.intercept_)
print("MSE:", mean_squared_error(y, y_pred))
```

```
Coefficients: [-1.0859654  0.0834224  0.03344602]
Intercept: 189.44529689100634
MSE: 435.9993195119308
```

Write a short reflection

1. Which model had the best performance
2. How did adding more features affect MSE
3. How can these models assist in early risk detection for heart patients

Answers:

1. Which model had the best performance?

The multiple linear regression model using age, trestbps, and chol performed better because it had better accuracy.

2. How did adding more features affect MSE?

Adding more features generally reduced the Mean Squared Error (MSE), indicating that the model's predictions became more accurate than using age alone.

3. How can these models assist in early risk detection for heart patients?

These models help estimate a patient's maximum heart rate based on clinical parameters, allowing healthcare professionals to identify potential heart health risks early and support timely diagnosis and treatment.

In []: